Multi-Hover-Point Relay Scheduling for UAV Emergency Delivery under a Fixed Transport Plan

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***Abstract*—Mountain floods damage roads and communication facilities at once, so ground vehicles can hardly enter the affected areas and unmanned aerial vehicles become a key means of delivery and communication support; severe relief, tight deadlines and the coupling between the two require a consistent physical model. The paper builds a "transport scheduling–communication support–resource allocation" chain solved in four sequential stages: the transport plan is fixed first and the central stage constructs a relay schedule meeting coverage, energy and resource constraints. Problem 1 is an energy-aware bin-packing model solved exactly by set-partitioning dynamic programming, giving an 18-trip plan at 59.24 kWh, 0.69% below greedy batching heuristics on that instance. Problem 2 adds heterogeneous fleet routing with time windows and recommends a 20-trip plan at 65.25 kWh satisfying all 31 hard deadlines. Problem 3 introduces a *multi-hover-point direct-transfer* relay mode: two relay aircraft cover all 11,713 relay-required samples on the 1-s grid at a relay energy of 5.43 kWh, whereas the tested single-hover heuristics attain only 51.0%, 62.7% and 82.1% with one, two and three airframes. Problem 4 compares $K{=}2$ and $K{=}3$ partitions and recommends $K{=}2$. A rebuild of the frozen relay plan reproduces the relay-required count within 0.3% and the relay energy within 0.1%; the weakest relay link keeps 0.38 dB of margin above -90 dBm.**

***Index Terms*—Air transportation, communications, emergency delivery, freight transportation and logistics, multi-hover-point relay scheduling, optimization and control**

I. Introduction

Mountain flood disasters strike suddenly and are spatially dispersed, often destroying roads and communication facilities at the same time. Continuous heavy rainfall may trigger flash floods, landslides and road collapses, isolating residents from stable ground transportation for a short period; in the early post-disaster stage, drinking water, emergency food, medical supplies and hygiene items must be delivered as soon as possible, while conventional vehicles can hardly enter the strongly undulating and severely damaged areas. UAVs are less sensitive to ground-road conditions and can serve as an important complementary means of emergency transportation.

In July 2026, torrential rainfall brought by typhoon "Maysak" caused severe flooding in Zhenlong Township, Hengzhou, Guangxi. Field reports indicated that flood water, rockfall and road collapse cut off access from several directions, leaving the core area temporarily without road traffic and with unstable communication; the township suffered simultaneous road, power and network outages and about 8,000 residents were trapped. During the response, food, drinking water and medicine had to be carried in on foot or dropped by air, and large UAVs also transported power-repair materials, while temporary communication and emergency power equipment were deployed to gradually restore contact. This episode shows that emergency delivery and communication support must be organized jointly under blocked roads, tight energy budgets and incomplete information.

From a technical viewpoint, completing delivery depends not only on whether a UAV can reach the target area, but also on payload, flight distance, climb height, along-route terrain clearance, energy reservation, deadlines and equipment turnover. A transportation UAV must also maintain command, telemetry and hand-over links during climb, cruise and descent; when fixed facilities are damaged or terrain blocks the ground gateway, airborne relay UAVs must provide temporary coverage at suitable positions and time windows. Material delivery under mountain floods therefore couples transportation, communication support and equipment-resource coordination.

The four stages studied here are ordered by dependency rather than by convention. Batching (Problem 1) fixes the indivisible unit loads and their per-area energy cost before any timing decision is made, because a route cannot be timed before its payload is known. Routing (Problem 2) then adds deadlines, airframe concurrency and battery turnover on top of these fixed batches. Relay scheduling (Problem 3) is meaningful only after the routing timeline exists, since the relay demand is exactly the set of trajectory points whose link to the gateway fails. Finally, resource partitioning (Problem 4) accounts peak concurrency on the joint plan, so it must come last. Each stage consumes only the outputs of its predecessors, which is what allows one consistent physical model to run through the chain without re-fitting. Transportation and communication are physically and temporally coupled—the relay demand is generated by the transport trajectories—but the chain solves them sequentially rather than jointly, so no joint optimum is claimed. Batching and routing form an input-generating module, resource partitioning is an extension analysis, and relay scheduling under the fixed transport plan is the core of this paper.

A. Related Work

Relay deployment under terrain blockage is critical for link quality: [1] showed that even a single well-placed relay can significantly improve link reliability, and in emergency

logistics [2] revealed a timeliness–energy trade-off. Cellular-connected delivery and joint UAV–ground routing have since been studied extensively: trajectory optimization for cellular-connected fleets [3], joint delivery routing of UAVs and ground vehicles from an urban-logistics perspective [4] and under regional restrictions [5], and multi-objective design of UAV transportation networks [6]. In cooperative mountain logistics, [7] planned drone–vehicle orchard transport, [8] developed three-dimensional path planning and scheduling for urban emergency UAVs under typhoon disasters, and [9] designed dedicated flood search-and-rescue UAVs. On the communication side, [10] optimized drone-to-everything links for medical services and [11] improved UAV-assisted downlink transmission, while [12] studied self-triggered cooperative control of tethered multi-UAV transportation under low communication, and [13] surveyed UAV–UGV collaborative sensing for low-altitude transportation. Heterogeneous multi-UAV swarms [14] and two-stage UAV facility location and scheduling with objective-robustness optimization [15] are also closely related. Most of these studies address either the transportation or the communication side: routing works rarely enforce continuous link coverage, while communication works rarely account for payload, deadline and battery-turnover limits. A few come closer: the siting model of [15] decides where ground facilities stand but not how airborne links are maintained during delivery; the sensing survey of [13] catalogs collaboration patterns without a transport–communication optimizer; [16] compares joint and decoupled multi-UAV path planning for search and rescue; and [17] designs delivery-UAV routes that reserve multi-hop relays. The closest precedent is the HDNF preprint [18] (an arXiv preprint submitted to IEEE, not yet peer reviewed), which jointly optimizes delivery task assignment, the 3-D deployment of hovering UAV base stations and C2-aware delivery path planning; its link model is a probabilistic elevation-angle LoS channel, its base stations hold fixed positions, and its objective trades C2 reliability against flight energy and deployment cost in synthetic simulations. These works, however, treat the relay as a static route constraint or a fixed-position asset rather than a scheduled resource: none decides the relay hover sequence, its transfer flights and link re-establishment, nor the turnover of a shared energy-component inventory over the transport timeline, and none verifies the links against terrain. The gap addressed here is therefore the sequential, physically consistent chain in which transport batching and routing are fixed first and the multi-hover-point relay schedule is then constructed under coverage, energy and resource constraints. Table I positions these works against the three capabilities the coupled problem requires simultaneously: continuous link coverage (L), payload/energy feasibility (P) and battery/component turnover (T).

TABLE I
POSITIONING OF THE RELATED WORK. L = CONTINUOUS LINK COVERAGE, P = PAYLOAD/DEADLINE FEASIBILITY, T = BATTERY/COMPONENT TURNOVER.

Stream	Refs.	L	P	T
Transport & delivery routing	[2]–[9]	–	✓	–
UAV communication & relay	[1], [10], [11], [16], [17]	✓	–	–
Platform, swarm & control	[12], [14]	–	–	–
Siting, sensing & surveys	[13], [15]	–	✓	–
Joint delivery & C2 (preprint)	[18]	✓	–	–
This work		✓	✓	✓

B. Contributions

The physical rules used throughout—nonlinear equivalent range, DEM-derived cruise altitude, climb energy, two-stage charging, link budget and fleet inventory—are prescribed by the relief scenario and adopted as given, not claimed as new. Within that scope:

- **Multi-hover-point relay scheduling (MHPRS), the core contribution.** We formalize a mode in which one relay UAV serves several hover points per take-off/return cycle, state the mode-dominance proposition (the single-hover feasible set is a subset) and characterize, for the test instance, the concurrent service windows that motivate the two-airframe plan. Two relay UAVs cover all 11,713 sampled relay-required points on the 1-s grid, against 51.0%, 62.7% and 82.1% for the tested single-hover heuristics with one, two and three airframes.
- **A consistent, cross-checked implementation.** Each headline number is recomputed from the frozen outputs by a separate script (coverage within 0.3%, relay energy within 0.1%); this is a reproducibility contribution, not a new physical law, and the sequential chain claims no joint optimum.
- **An energy-feasibility discretization for batching.** The batching subproblem itself is solved by a standard subset enumeration plus set-partitioning DP; the added element is the reformulation of the continuous energy constraint into feasible pair loads, giving a verified benchmark (18 trips, 59.24 kWh, 0.69% below FFD/BFD); no DP novelty is claimed.

II. System Model and Problem Formulation

A. System Overview

The temporary dispatch center O_{01} provides ground control, the communication gateway (G01, antenna 20 m above ground), and battery/energy-component management. Three types of transportation UAVs (A/B/C) execute round trips under a return-safety margin $\rho = 20\%$, delivering box by box to 15 service areas subject to payload, volume and energy constraints. Shadowed back-slope segments are relayed by two relay UAVs using multi-hover-point direct transfer, maintaining end-to-end connectivity through a "transport–relay–G01" two-hop link (both directions ≥ -90 dBm). Relay UAVs return to base to swap energy components after each take-off/service/return cycle. Fig. 1 summarizes the overall architecture.

Model assumptions. Five assumptions define the scope, each trading some realism for reproducibility. (A1) Terrain and link state are static within each one-second sample, matching the verification grid; shorter outages are therefore outside the coverage guarantee. (A2) Terrain occlusion is binary with a fixed 10 dB loss rather than a diffraction model, which makes the line-of-sight test deterministic and fully reproducible but less radio-faithful. (A3) Wind, precipitation

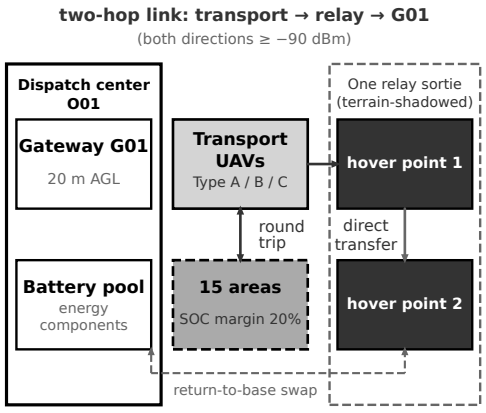


Fig. 1. UAV transportation–communication collaborative system architecture.

and equipment faults are excluded, so plans are valid for a single deterministic post-disaster scenario. (A4) Demand (box masses, volumes and deadlines) is known and fixed; stochastic or rolling demand is left to future work. (A5) Relay demand is computed from the planned transport trajectories, so the communication problem is conditional on the Problem-2 schedule; this preserves the stage ordering but excludes replanning after unplanned detours.

B. Shared Physical Model (used by all stages)

1) Distance and cruise altitude: The great-circle distance between O_{01} and service area S_j uses the Haversine formula:

$$\begin{aligned} d_{Oj} &= 2R \arctan\sqrt{\frac{a}{1-a}}, \\ a &= \sin^2\frac{\Delta\varphi}{2} + \cos\varphi_1 \cos\varphi_2 \sin^2\frac{\Delta\lambda}{2}, \end{aligned} \tag{1}$$

where $\Delta\varphi, \Delta\lambda$ are latitude/longitude differences and R is the Earth radius. The cruise altitude along a segment takes the DEM elevation by the nearest-pixel rule plus 50 m:

$$h^{\text{cruise}}_{(j)} = h^{\max}_{(j)} + 50. \tag{2}$$

2) Trip energy and return-safety margin: The total round-trip energy consists of four parts (outbound level, return level, outbound climb, return climb):

$$E^{\text{total}}_{(i,j)}(q) = E^{\text{go}}_{\text{hor}}(q) + E^{\text{ret}}_{\text{hor}}(0) + E^{\text{go}}_{\text{climb}}(q) + E^{\text{ret}}_{\text{climb}}, \tag{3}$$

with outbound payload q and empty return. The level-flight parts follow a nonlinear equivalent-range law: with L_0 the empty and L_F the full-load standard range and Q the rated payload, the equivalent range at payload q is

$$L_g(q) = L_0 - (L_0 - L_F)\Big(\frac{q}{Q}\Big)^{3/2}, \tag{4}$$

$$E^{\text{go}}_{\text{hor}}(q) = E^{\text{bat}}\frac{d}{L_g(q)}, \qquad E^{\text{ret}}_{\text{hor}}(0) = E^{\text{bat}}\frac{d}{L_0}, \tag{5}$$

so that loading reduces range superlinearly. The climb parts are potential energy terms with efficiency $\eta_c = 0.72$ and no descent recovery:

$$E^{\text{go}}_{\text{climb}}(q) = \frac{g(m_0+q)(H-h_O)}{\eta_c \cdot 3.6\times 10^6}, \tag{6}$$

$$E^{\text{ret}}_{\text{climb}} = \frac{g\, m_0 (H - h_S - 30)}{\eta_c \cdot 3.6\times 10^6}, \tag{7}$$

TABLE II
MAIN NOTATION.

Symbol	Meaning	Unit
S_j / O_{01}	service area / dispatch center	–
n_j	number of boxes at S_j	–
μ_k	UAV type of trip k ($\in \{A,B,C\}$)	–
m^{safe}	maximum safe payload	kg
ρ	return-safety margin	–
N_{trips}	number of round trips	–
T_{total}	cumulative operating time	s
$E_{\text{hor}}, E_{\text{climb}}, E_{\text{total}}$	level/climb/total energy	kWh
$L_g(q)$	nonlinear equivalent range	m
L_0, L_F	empty / full-load standard range	m
m_0, Q	curb mass / rated payload	kg
v_{up}, v_{dn}	climb / descent speed	m/s
η_c	climb energy efficiency (0.72)	–
h^{cruise}	cruise altitude (max elev. $+50$ m)	m
SOC	state of charge (return $\geq 20\%$)	–
$C_{\max}$	mission completion time (latest return)	s
L_{soft}	accumulated soft delay	min
P_r	received power (threshold -90 dBm)	dBm
$\mathcal{P} = \{P_1, \ldots, P_K\}$	task partition (K=2 or 3)	–
$N(P_i, q)$	peak concurrent demand of q	units
D, G, J	scale/gap/balance	mixed
I_q	resource inventory vector	units

where m_0 is the curb mass, H the cruise altitude, h_O the depot elevation, h_S the service-area elevation and the 30 m offset is the hand-over height above the delivery point; energies are in kWh. The usable energy with margin ρ is

$$E^{\text{avail}}_{(i)} = E^{\text{bat}}_{(i)} \cdot (1-\rho), \qquad E^{\text{total}}_{(i,j)}(m^{\text{safe}}_{(i,j)}) \leq E^{\text{avail}}_{(i)}, \tag{8}$$

from which the maximum safe payload $m^{\text{safe}}_{(i,j)}$ is obtained by binary search. The corresponding trip time combines vertical and horizontal flight with ground handling,

$$\begin{aligned} t_{\text{trip}} = (2H - h_O - h_S - 30)\Big(\frac{1}{v_{\text{up}}} + \frac{1}{v_{\text{dn}}}\Big) + \frac{2d}{v} \\ + t_{\text{prep}} + t^0_{\text{hand}} + n\,(t_{\text{load}} + t^1_{\text{hand}}), \end{aligned} \tag{9}$$

where n is the number of boxes on the trip, $v_{\text{up}}, v_{\text{dn}}$ are the climb and descent speeds and the last terms are the preparation, loading and hand-over times of Table III.

C. Notation

Table II lists the main notation; unless stated otherwise, units follow the problem data (length in m, mass in kg, time in s, energy in kWh and power in dBm).

III. METHODOLOGY

A. Overview of the Unified Framework

We establish a "transport–communication–resource" collaborative optimization chain: with 30 m DEM terrain, 15 service areas, 80 indivisible boxes and heterogeneous UAV fleets plus link parameters as unified inputs, we successively build DEM-ELBPP (Problem 1), HF-VRPBT (Problem 2), MHPRS (Problem 3) and SC-MAE-RG (Problem 4). The four layers introduce return-safety margin and batching, deadline concurrency and battery turnover, full-process continuous communication, and partitioning resource allocation, respectively; the resulting four-stage sequential framework is shown in Fig. 2.

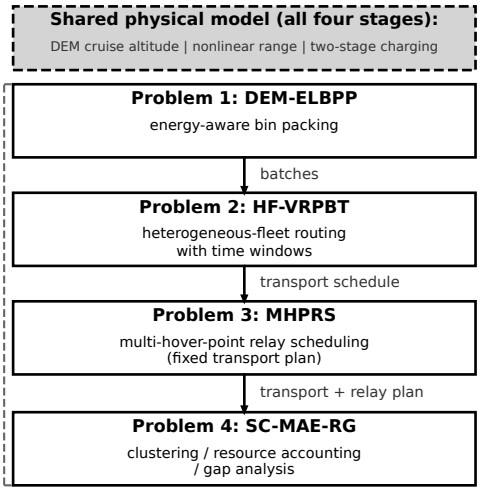


Fig. 2. Four-stage sequential optimization framework.

B. Problem 1: DEM-Driven Energy-Aware Bin Packing (DEM-ELBPP)

For the box set B_j at service area S_j ($|B_j| = n_j$), batching is modeled as an energy-constrained hierarchical bin packing: each trip π_k loads a nonempty subset of B_j executed by type $\mu_k \in \{A, B, C\}$, subject to payload, volume and energy constraints. Among the three objectives (trip count, total energy, cumulative operating time) we adopt *lexicographic* optimization instead of a weighted sum, because the three have different dimensions and arbitrary weights would change the optimum subjectively:

$$\min_{\text{lex}} \Big(N_{\text{trips}},\ \sum_k E_k,\ T_{\text{total}} \Big) \tag{10}$$

subject to

$$\bigcup_k \pi_k = B_j, \qquad \pi_k \cap \pi_{k'} = \varnothing\ (k \neq k'), \tag{11}$$

$$\sum_{b \in \pi_k} m_b \le m^{\text{safe}}_{(i,j)}, \qquad \sum_{b \in \pi_k} V_b \le V^{\max}_{(i)}, \tag{12}$$

$$\mu_k \in \{A, B, C\}, \quad \pi_k \subseteq B_j. \tag{13}$$

The key idea is to first back-solve the safe payload from the energy equation and then perform set partitioning under the payload–volume–energy triple constraint, discretizing the continuous energy constraint into bin-packing feasibility.

Candidate batches are enumerated per area (the largest area holds $n_j = 15$ boxes, so at most $2^{15} - 1 = 32767$ subsets), and the exact solution uses set-partitioning dynamic programming:

$$\begin{aligned} \mathrm{dp}[\text{mask} \cup \pi] = \min \big(& \mathrm{dp}[\text{mask} \cup \pi], \\ & \mathrm{dp}[\text{mask}] \oplus (\pi) \big), \end{aligned} \tag{14}$$

where $\oplus$ is vector addition and triples are compared lexicographically; the optimal batch sequence is then recovered by backtracking the predecessor chain. Two implementation facts bound the computational cost. First, the 45 machine–area safe payloads are precomputed once by a 60-iteration binary search on the energy equation, so the DP never evaluates the continuous energy model. Second, because each area holds at most fifteen boxes, the filtered candidate count stays far below the 2^{15} masks of that area; mixed-type batching within an area is allowed, so the DP chooses the cheapest machine per candidate subset rather than fixing the type in advance. The overall procedure is summarized in Algorithm 1.

Algorithm 1 Exact batching (subset enumeration + set-partitioning DP)

1: **input** box sets B_j, UAV parameters, DEM, margin ρ
2: binary-search $m^{\text{safe}}_{(i,j)}$ for each machine/area
3: enumerate feasible subsets per area, filter by payload/volume/energy $\rightarrow$ candidate batches C_j
4: init $\mathrm{dp}[0] = (0, 0, 0)$, others ∞; relax candidate batches lexicographically
5: backtrack to output the optimal batching plan
6: scan $\rho = 20\% \sim 40\%$ and repeat for sensitivity
7: **output** optimal batching plan, per-area machine/energy, margin sensitivity

C. Problem 2: Heterogeneous Fleet Routing with Battery and Time Windows (HF-VRPBT)

Problem 2 relaxes the single-area structure and adds resource and time-window constraints: each trip may visit one or more service areas before returning. The decisions jointly determine box batching, visit sequence, machine type μ_k, the executing airframe, the shared battery and the trip start time. With B the 80 boxes, B^{hard} the 31 hard-deadline boxes and B^{soft} the rest, the constraints enforce one delivery per box, per-trip payload and volume limits, segment-wise residual payload, trip energy, return SOC $\ge \rho$, hard deadlines $t_b \le d_b$ for all $b \in B^{\text{hard}}$, airframe and battery mutual exclusion, and recharge to 100% before battery reuse. Formally, with $x_{kb} \in \{0, 1\}$ marking the assignment of box b to trip k, $[s_k, r_k]$ the interval trip k occupies its airframe (s_k: preparation start, r_k: return) and $[s_k, c_k]$ the interval its battery is occupied ($c_k = r_k + T_{\text{chg}}(\mathrm{SOC}_k)$: fully recharged), the scheduler accepts a plan only if

$$\begin{aligned} & \textstyle\sum_k x_{kb} = 1 \ \ (b \in B), \qquad \sum_b x_{kb} m_b \le m^{\text{safe}}_{(\mu_k)}, \\ & \textstyle\sum_b x_{kb} V_b \le V^{\max}_{(\mu_k)}, \qquad t_b \le d_b \ \ (b \in B^{\text{hard}}), \\ & \mathrm{SOC}_k = 1 - E_k / E^{\text{bat}}_{(\mu_k)} \ge \rho, \\ & [s_k, r_k] \cap [s_{k'}, r_{k'}] = \varnothing \ \ \text{(same airframe)}, \\ & [s_k, c_k] \cap [s_{k'}, c_{k'}] = \varnothing \ \ \text{(same battery)}, \end{aligned} \tag{15}$$

i.e. trips sharing an airframe or a battery never overlap, and a battery is reusable only after a full recharge, which takes $T_{\text{chg}}(\mathrm{SOC}) = T^{\text{full}}_{(\mu)} [0.35 + 0.65\,(0.9 - \mathrm{SOC})/0.9]$ s under the two-stage charging rule ($T^{\text{full}} = 1800/2400/3000$ s for A/B/C; all return SOCs in the computed plans lie below 0.9). Along the visit sequence π_k the residual payload is recursive, $\Lambda_{k,j} = \Lambda_{k,j-1} - \sum_{b:\mathrm{pos}(b)=j} m_b$, $\Lambda_{k,0} = \sum_b x_{kb} m_b$: each leg carries exactly the boxes still undelivered; multi-area sequences are restricted to type C, and the delivery instant t_b follows the cumulative leg-time recursion (climb–cruise–descend per leg at the DEM-checked cruise altitude). Four objectives—timeliness, makespan $C_{\max}$ (latest return), transport energy and trip count—are compared in a non-dominated sense, with the soft delay $L_{\text{soft}} = \sum_{b \in B^{\text{soft}}} \max(0, t_b - d_b)$ measuring the lateness of the remaining boxes. Since the full

trip-routing space is intractable, we enumerate a restricted candidate set (whole-area trips inherited from Problem 1, split first-batch tasks and short cross-area tours) and apply a deterministic urgency-first greedy schedule that inserts trips by nearest hard deadline. Inherited whole-area trips preserve the energy-optimality of the Problem-1 batches, split first-batch tasks decouple urgent first deliveries from the residual load so that hard deadlines are met before the full batch is ready, and short cross-area tours absorb small residual loads that would otherwise each cost a dedicated round trip; each insertion recomputes the occupancies of the chosen airframe and battery and rejects placements violating the conditions above. The search evaluates 57 candidate plans and yields five non-dominated plans (Table VI): 19 trips (66.30 kWh), the recommended 20-trip plan (65.25 kWh, $C_{\max} = 2.72$ h, soft delay 137.6 min), two further 20-trip plans that trade a longer makespan for smaller soft delay, and 21 trips (66.00 kWh); the recommended 20-trip plan dominates on energy and completion time, uses all eight airframes and satisfies all 31 hard-deadline boxes.

D. Problem 3: Task-Driven Multi-Hover-Point Relay Scheduling (MHPRS)

1) Link feasibility: Link feasibility uses a free-space path-loss model with a binary DEM line-of-sight occlusion test (10 dB occlusion loss when blocked):

$$P_r = P_t + G_t + G_r - 32.45 - 20\log_{10} f - 20\log_{10} d - L_{\text{obs}} - L_{\text{sys}}, \tag{16}$$

with $L_{\text{obs}} \in \{0, 10\}$ dB and $L_{\text{sys}} = 3$ dB; f is in MHz, the three-dimensional slant range d is in km, powers are in dBm and antenna gains in dBi. Both directions (transport↔relay and relay↔G01) must satisfy $P_r \geq P_{\text{thr}} = -90$ dBm. The relay platform is a 23.5 kg multi-rotor with a 3.2 kWh swappable energy component: it cruises at 1.15 kW, and draws a flat $1.05 + 0.05$ kW over each service window (1.05 kW platform hover plus 0.05 kW communication payload), recharges in $T_{\text{chg}} = 500 + 1300\,(1 - \text{SOC})$ s under the two-stage charging rule (1800 s from a fully depleted component), and must land with SOC ≥ 0.20; six components circulate between the two airframes and the chargers.

2) Multi-hover-point relay scheduling (MHPRS): **Definition (MHPRS).** Given a set of relay-required time intervals and a set of candidate hover points, a *relay sortie* is a take-off/service/return cycle of one relay UAV serving an ordered sequence of hover points connected by direct-transfer flights; MHPRS is to cover all relay-required points by the minimum-cost set of sorties under link-budget, hover-height, resource-exclusion, return-SOC and component-inventory constraints. Classical relay placement installs fixed relays so that a static set of terminals becomes connected and optimizes positions only. MHPRS differs in three respects: (i) the demand is *time-varying*, since relay-required points appear and disappear as the transport trips progress, so coverage is spatio-temporal rather than purely spatial; (ii) one airframe may serve an *ordered sequence* of hover points within one sortie, and the transfer flight and link re-establishment between consecutive points consume time and energy that enter the constraints; and (iii) only two airframes share six energy components, so hover-height, energy and component-turnover limits bind the solution. Formally, the constraints comprise full-process coverage (each transport point is either directly connected to G01 or lies in some sortie's service interval with both hops above threshold), relay-before-task $s_c \leq \min\{t_p\}$, bidirectional link budget $P_r \geq P_{\text{thr}}$, hover-height limit $h_c \leq h_c^{\text{terr}} + 300$ m, resource exclusion, return SOC ≥ 0.20, component inventory (in-flight plus charging ≤ 6, reusable only when fully charged), coverage rate $\geq \eta_{\text{target}}$, and the transfer–link-establish timing between consecutive segments; the reported coverage counts logged relay-required sample rows, including boundary duplicates between consecutive segments.

The solution procedure mirrors this structure: relay-required points are clustered into service intervals by greater-than-400 s gaps; candidate hover points are generated on the 0.01° DEM grid within 300 m above ground and 12 km of the depot, keeping only points from which every relay-required point of a segment passes the link budget; a greedy pass selects the fewest candidates per interval, splitting at the median time when a segment remains uncovered, and the resulting adjacent segments are packed into sorties under the component-inventory and return-SOC constraints with earliest-available assignment of the two airframes.

Proposition 1 (mode dominance, elementary). Let $\mathcal{S}_{\text{SH}}$ be the set of coverage plans feasible when every sortie is restricted to a single hover point, and $\mathcal{S}_{\text{MH}}$ the set feasible under MHPRS with ordered hover-point chains. Then $\mathcal{S}_{\text{SH}} \subseteq \mathcal{S}_{\text{MH}}$; consequently, for any coverage target η, (i) if η is attainable at all by the single-hover mode with a given fleet, it is attainable by MHPRS with the same fleet, and (ii) the minimum energy required under MHPRS is no greater than under the single-hover mode. *Proof.* A single-hover sortie is the length-one special case of an MHPRS chain: relabeling every single-hover sortie as a length-one chain adds no transfer flight, and all other constraints (link budget, hover height, resource exclusion, return SOC, component inventory) are identical, so the relabeled plan remains feasible at the same energy. Minimizing a common cost over a superset of feasible plans cannot raise the optimum. □

Empirical gap. The tested single-hover heuristics attain only 51.0%, 62.7% and 82.1% with one, two and three airframes, whereas MHPRS covers all 11,713 sampled relay-required points with four sorties.

Observation 1 (relay concurrency on the test instance). No feasible single-airframe schedule was found in the evaluated search. Two first-segment service windows overlap, $[751.8, 1512.6]$ s and $[752.8, 1712.7]$ s, sharing 759.8 s of time, while their point-cloud centroids lie 7188.5 m apart. The union of their relay-required points contains 3765 samples; of the 397 enumerated candidate hover points the best covers 2567, and the single-airframe schedule search returns no feasible plan. The two-airframe plan is therefore used. This requirement is empirical: it rests on exhaustive enumeration over the evaluated candidate pool and on the failure of the single-airframe search, not on a geometric lower bound, because an MHPRS airframe may relocate between hover points.

Algorithm 2 Task-driven multi-hover-point relay assignment (MHPRS)

1: **input** Problem-2 schedule, DEM, link params, 2 relay UAVs, 6 components
2: sample trajectories per second, mark relay-required points
3: cluster relay-required points by time (>400 s gap) into service intervals
4: generate candidate hover points (DEM-covered, $\leq$300 m, $\leq$12 km)
5: greedy select points per cluster; split by median time if uncovered
6: pack adjacent segments into sorties (multi-hover direct transfer); earliest-available assignment
7: check component turnover (two-stage charging) and return SOC
8: full verification by per-second sampling and boundary checks
9: **output** relay schedule: hover-point chains, service windows, verified coverage and per-sortie SOC

The combined stage-3 objective is

$$\begin{aligned}\min Z &= E_{\text{trans}} + E_{\text{relay}},\\ E_{\text{relay}} &= \sum_{c} \Big(E_c^{\text{fly}} + (P_{\text{hover}} + P_{\text{comm}})\tfrac{e_c - s_c}{3600} \\ &\qquad + P_{\text{hover}}\tfrac{w_c}{3600} \Big),\end{aligned} \tag{17}$$

where w_c is the accumulated hovering wait time within sortie c and E_c^{fly} is the outbound/transfer/return flight energy. The three terms are computed over mutually disjoint intervals of the sortie timeline: inter-segment transfer time is charged to E_c^{fly} and excluded from the served-hover window $[s_c, e_c]$, so no energy is double-counted. The stage-3 objective is subject to the constraint set listed in the MHPRS definition above. The stage-3 solver is a greedy construction over the candidate hover points, so the returned schedule satisfies the constraints above but minimizes neither the relay energy nor the sortie count in a proven sense; Observation 1 supports the two-airframe plan only within the evaluated candidate pool. The resulting multi-hover-point direct-transfer mode, together with its coverage advantage over the single-hover baseline, is illustrated in Fig. 3 and summarized in Algorithm 2.

E. Problem 4: Partition-Based Resource Accounting (SC-MAE-RG)

Problem 4 takes the Problem-3 joint plan as input and groups the 15 service areas into $K \in \{2, 3\}$ task groups, subject to each area belonging to exactly one group, each group containing at least three areas (an explicit operational grouping assumption: one- or two-area groups degenerate the grouping into per-area accounting; the one-area variant is reported in the benchmark for completeness), and any areas that share a transport trip being placed in the same group. Each group independently accounts for the eight resource classes (A/B/C airframes, A/B/C batteries, relay airframes and energy components) at their peak concurrency, because resources may not be reassigned across groups. For group P_i and resource class q the peak concurrent demand is

$$\begin{aligned}N(P_i, q) = \max_{\tau} \big| \{\, u : S_u \cap P_i \neq \varnothing,\ \tau \in [a_u, b_u], \\ \text{type}(u) = q \,\} \big|,\end{aligned} \tag{18}$$

from which the configuration scale $D = \sum_i \sum_q N(P_i, q)$, the inventory gap $G = \sum_q \max(0, \sum_i N(P_i, q) - I_q)$, the cross-group relay count, the replication total and the inter-group balance index J follow; both D and G sum the eight resource classes with equal weight, so they act as inventory-count proxies rather than cost-weighted measures. Candidate partitions are enumerated from 80 K-means seeds on standardized geographic coordinates (with the S008–S011 shared-sortie coupling enforced), two hand-built skeletons and the relay-service-induced split, and ranked lexicographically by (unique cross-group relays, extra cross-group replicas, D, G, J); the winner is the best partition within this pool, not a certified optimum over all partitions. The balance index is the mean coefficient of variation across six workload dimensions (boxes, mass, sorties, energy, busy duration and relay count),

$$J(\mathcal{P}) = \frac{1}{6} \sum_{\ell=1}^{6} \frac{\sigma_\ell(\mathcal{P})}{\mu_\ell(\mathcal{P})}, \tag{19}$$

so that J=0 marks a perfectly even split and larger values indicate workload concentration. On the test instance the recommended K=2 partition $[11, 4]$ attains D=31 and G=5, against D=38 and G=10 for K=3 $[7, 4, 4]$, so K=2 is preferred (Sec. IV details the per-class gaps and the zero-replication control). An entropy-weighted TOPSIS ranking (Sec. IV) identifies S001 as the most delivery-pressured area, consistent with the heavy load carried by the larger group.

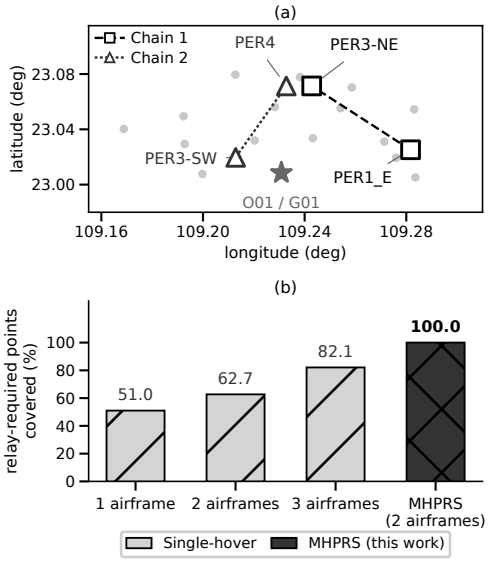


Fig. 3. Multi-hover-point direct-transfer relay scheduling mode. (a) Hover-point chains of the two multi-hover sorties on the real node coordinates: Chain 1 is sortie R01M1 and Chain 2 is sortie R02M2; gray dots are the 15 service areas, the star is the dispatch center O01/G01, and each transfer leg is the direct straight flight between successive hover points; (b) coverage of the 11,713 sampled relay-required points versus the number of relay airframes under the single-hover round-trip baseline, compared with the proposed MHPRS.

TABLE III
HETEROGENEOUS FLEET AND ENERGY PARAMETERS OF THE TEST INSTANCE.

Parameter	Type A	Type B	Type C
Payload Q (kg)	25	30	80
Curb mass m_0 (kg)	70.0	65.0	69.9
Cabin volume (m^3)	0.06	0.073	0.25
Cruise speed v (m/s)	12	15	15
Climb/descent speed $v_{\mathrm{up}}/v_{\mathrm{dn}}$ (m/s)	3.0/2.5	3.0/2.5	2.5/2.0
Battery capacity E^{bat} (kWh)	4.5	4.0	8.0
Range empty/full L_0/L_F (km)	25/20	28/16	26/12
Airframes	4	2	2
Shared batteries	6	4	4
Full-charge time (s)	1800	2400	3000
Preparation t_{prep} (s)	300	300	300
Loading per box t_{load} (s)	30	30	30
Hand-over base + per box (s)	150+30	150+30	180+36

TABLE IV
COMMUNICATION AND TERRAIN PARAMETERS OF THE TEST INSTANCE.

Parameter	Value
Carrier frequency	2400 MHz
DEM resolution	30 m
Occlusion loss (blocked line of sight)	10 dB
System loss L_{sys}	3 dB
Receiver sensitivity / design margin	−98 dBm / 8 dB
Receiving threshold P_{thr}	−90 dBm
Transport UAV transmit power / antenna gain	20 dBm / 3 dBi
Relay access link (to transport UAVs)	20 dBm / 6 dBi
Relay backhaul link (to G01)	19 dBm / 8 dBi
Gateway G01 (20 m mast)	27 dBm / 12 dBi

IV. EXPERIMENTAL DESIGN AND RESULTS

A. Setup

All results are computed on the standard mountainous test instance supplied with the problem: one dispatch center O_{01} (gateway G01), 15 service areas, 80 indivisible boxes of four categories, three transport UAV types with a shared battery pool, two relay UAVs with six energy components, a 30 m DEM and the specified link parameters. The exact batching solver is compared with greedy FFD and BFD over N=30 randomized instances derived from the real case, and the relay plan with single-area, single-hover-point and augmented-fleet controls. Metrics are trips, total energy, makespan, soft delay and coverage; all solvers use Python (NumPy/pandas) and the values are reproduced by an independent verification script. Relay-required points are counted on per-second sampled trajectories (1 s step), to which the main-plan coverage figures refer, whereas the threshold and clearance scans resample the same trajectories at a coarser 5 s search step and therefore report fewer points. Table III lists the fleet and energy parameters and Table IV the communication and terrain parameters; the return-SOC floor is $\rho = 20\%$ and the climb efficiency is 0.72. The four box categories are medical kits (3 kg, 0.012 m^3), drinking water (14 kg, 0.027 m^3), emergency food (8 kg, 0.028 m^3) and hygiene items (6 kg, 0.035 m^3).

B. Baseline Comparison

Table V compares the exact dynamic-programming batching with the classical FFD, BFD and WFD heuristics over N=30 randomized instances. On average the proposed method returns the fewest trips and the lowest energy (67.18 kWh), and the energy penalty of the heuristics—2.49% as the relative increase of the mean energy, 2.30% as the mean of the per-instance increases—is significant ($p < 10^{-4}$, two-sided Wilcoxon signed-rank test). On the real instance the same ordering holds with a much smaller gap (59.24 kWh against 59.65 kWh, i.e. 0.69%), so the exact solver's advantage lies in consistent behaviour across instances rather than in a single large energy margin. The instances use the fixed seed 20251015: the box count is uniform on $\{40, \ldots, 120\}$, the service-area coordinates are jittered with $\sigma = 0.0005°$, and the categories are resampled from the real case. All 30 generated instances were accepted and none discarded. The family perturbs only batching-stage inputs (box count, category mix, local coordinates) around the same 15 areas, so it is a batching-subproblem benchmark, not an end-to-end geographic validation; routing, relay and partitioning are evaluated on the real instance only. The Wilcoxon test on the 29 non-zero pairs gives a two-sided $p = 2.56 \times 10^{-6}$ for FFD and BFD under the normal approximation (an explicit exact enumeration gives $p = 3.7 \times 10^{-9}$, so the significance conclusion is unchanged either way), and the per-instance energy spans 42.6–89.3 kWh with 14–22 trips, so the comparison covers instances both smaller and larger than the real case.

TABLE V
COMPARISON OF THE PROPOSED EXACT DYNAMIC-PROGRAMMING GROUPING AGAINST CLASSICAL BIN-PACKING HEURISTICS OVER $N = 30$ RANDOMIZED INSTANCES DERIVED FROM THE REAL CASE. SORTIES AND TOTAL ENERGY (IN KWH) ARE REPORTED AS MEAN±STD; THE IMPROVEMENT IS THE RELATIVE INCREASE OF THE MEAN ENERGY, $(\bar{E}_h - \bar{E}_p)/\bar{E}_p$ (THE MEAN OF THE PER-INSTANCE INCREASES IS 2.30% FOR FFD/BFD AND 2.63% FOR WFD); p-VALUES ARE FROM A TWO-SIDED WILCOXON SIGNED-RANK TEST ON PER-INSTANCE TOTAL ENERGY (PROPOSED VS. EACH HEURISTIC).

Method	Sorties	Energy	Improv.	p-value
Proposed (Exact DP)	18.5±2.3	67.18±14.32	—	—
FFD	18.6±2.4	68.85±15.26	+2.49%	$< 10^{-4}$
BFD	18.6±2.4	68.85±15.26	+2.49%	$< 10^{-4}$
WFD (worst-fit)	18.5±2.3	69.06±15.23	+2.79%	$< 10^{-4}$

C. Ablation and Sensitivity

The models are examined by ablation and one-factor sensitivity analysis.

For the routing stage, the candidate search evaluates 57 plans, of which five are non-dominated (Table VI). The recommended 20-trip main plan has the lowest makespan (2.72 h) and near-minimal energy (65.25 kWh); its soft delay of 137.6 min concerns only soft-deadline boxes, because all 31 hard-deadline boxes are on time. By aircraft type it flies four A trips (9.59 kWh), seven B trips (16.32 kWh) and nine C trips (39.34 kWh), the C type carrying 60.3% of the transport energy; all eight airframes are used, the four A airframes flying one trip each and the two B and two C airframes up to four and five. Nineteen trips serve one area and only T18 chains two (S011 then S008), so cross-area chaining pays off only where the inherited batches leave small residual loads. The makespan is set by the last C trip to S007, returning at 9794 s.

The recommended plan is taken forward as the transport input of the relay stage. Because the relay schedule is con-

TABLE VI
NON-DOMINATED ROUTING PLANS (FIVE OUT OF 57 EVALUATED CANDIDATES). THE RECOMMENDED MAIN PLAN IS MARKED IN BOLD.

Trips	Energy (kWh)	Makespan (h)	Soft delay (min)
19	66.30	2.94	105.77
20	**65.25**	**2.72**	**137.61**
20	65.57	2.80	109.95
20	67.05	2.80	97.56
21	66.00	2.80	95.50

TABLE VII
RETURN-SAFETY MARGIN SCAN FOR THE BATCHING STAGE (PROBLEM 1).

ρ	Trips	Energy (kWh)	Duration (h)	Remark
10%–20%	18	59.24	9.11	recommended ρ=20%
25%	19	61.19	9.61	
30%	20	67.36	10.17	
35%	25	78.19	12.65	
40%	—	—	—	infeasible (S004, S008)

ditional on the transport timeline, the relay demand below is identified and verified point-by-point for this urgency-first plan only, and the relay results are scoped accordingly.

For the batching model, forcing a single aircraft type raises the energy to 112.39 kWh (all A), 68.48 kWh (all B) or 74.06 kWh (all C) against 59.24 kWh for the mixed plan, so the aircraft–payload–range match dominates. In the exact plan twelve areas need one trip and three need two (S001–S003); the binding constraint is the C trip to S004, returning at SOC 0.228, and the cruise altitudes of the 18 trips span 273–592 m. Table VII scans the return margin: the 18-trip plan is unchanged for $\rho \leq 20\%$ and becomes infeasible at ρ=40%, where the empty round trips to S004 and S008 exceed the equivalent range.

For the relay stage, Table IX tightens the receiving threshold: from -98 to -95 to -90 dBm the relay-required points grow from 0 to 806 to 2311 and the directly served areas shrink from 15 to 10 to 3. The 5-s scan at -90 dBm contains these 2311 relay-required samples, whereas the 1-s verification set contains 11,713 logged rows; on the latter set the tested single-hover heuristics cover 51.0%, 62.7% and 82.1% with one, two and three airframes, so they fall short of full coverage. Raising the safety clearance above the terrain-following cruise altitude from 0 to 25 to 50 m increases the direct-connection rate only moderately, from 61.7% to 63.2% to 64.7%, so the relay requirement is governed mainly by the receiving threshold rather than by the clearance margin. The rebuilt four-sortie plan assigns 5,340, 508, 1,383 and 4,482 relay-required points to the four sorties, with a rebuilt relay energy of 5.432 kWh against the frozen 5.433 kWh (a 0.02% deviation), and the weakest verified link measures -89.62 dBm, 0.38 dB above the -90 dBm threshold. Table VIII breaks the verification down per sortie: the two multi-hover sorties (R01M1 visiting two distinct hover points over three service windows, R02M2 visiting two) carry 83.9% of the relay-required points, and their rebuilt energies match the frozen log within 1.2% even though one transfer hover point of each is absent from the frozen log and had to be re-searched independently; the largest per-sortie deviation (-3.3% on the single-hover R02M1) stems from the prior climb-efficiency value being higher than the one implicit in the frozen log, not from hover-point uncertainty. The reconstruction rule is conservative: the canonical demand set contains $11{,}713$ logged one-second rows with 68 boundary duplicates ($11{,}645$ unique sortie–time pairs), and the two missing transfer hover points were re-searched on the 0.01° DEM grid within 300 m above ground and 12 km of the depot, taking among all candidates that cover the whole segment the one nearest to the segment's point-cloud centroid. Because two hover points are reconstructed from the DEM grid rather than read from the frozen log, this recomputation is a reconstruction check of the frozen plan rather than a fully independent measurement; it uses the design parameters unchanged, including the transfer-climb efficiency $\eta_c = 0.72$, and leaves the frozen plan itself unaffected.

TABLE VIII
PER-SORTIE VERIFICATION OF THE FOUR-SORTIE RELAY PLAN. POINTS ARE LOGGED ONE-SECOND SAMPLES, INCLUDING 68 BOUNDARY DUPLICATES (11,645 UNIQUE SORTIE–TIME PAIRS); ENERGY IS THE REBUILT VALUE AGAINST THE FROZEN LOG (KWH); SOC IS THE REBUILT RETURN STATE OF CHARGE; (×2): THE SAME PHYSICAL HOVER POINT SERVES TWO SERVICE WINDOWS.

Sortie	Hover-point chain	Points	Energy (kWh)	SOC
R01M1	PER1_E (×2) → PER3_NE	5340	1.821 / 1.823	0.431
R01M2	single hover point	508	1.059 / 1.056	0.669
R02M1	single hover point	1383	0.678 / 0.701	0.788
R02M2	PER3_SW→PER4	4482	1.874 / 1.853	0.414
Total	6 hover points, 7 windows	11713	5.432 / 5.433	–

TABLE IX
RECEIVING-THRESHOLD SCAN FOR THE RELAY STAGE (5 S SEARCH SAMPLING, 6028 SAMPLED TRAJECTORY POINTS).

Threshold	Relay-required	Direct-connected	Direct-served areas
-90 dBm	2311	3717	3
-95 dBm	806	5222	10
-98 dBm	0	6028	15

For the partitioning stage, the K=2 plan groups S001–S009, S011 and S015 (11 areas, 68 boxes, 15 sorties, 54.37 kWh) against S010 and S012–S014 (4 areas, 12 boxes, 5 sorties, 10.87 kWh), with scale D=31, gap G=5 and balance index J=0.613; the K=3 plan splits $9/6/5$ sorties ($35.87/18.50/10.87$ kWh) with D=38, G=10 and J=0.438. The zero-replication control lowers the scale to D=30 and D=31 respectively, isolating cross-group relay replication as the cause of the extra redundancy. The entropy-weighted TOPSIS ranking behind the S001 identification measures *delivery pressure*—how heavily an area loads the transport plan—rather than physical vulnerability: the weights are 0.412 (mass), 0.385 (volume), 0.106 (distance), 0.097 (elevation), so the index is demand-dominated. Table XII lists the five most-pressured areas: S001 ranks first, combining the largest demand ($1.9\times$ the next-largest area) with the shortest distance, the natural first wave. Table XI details the peak concurrent demand of the recommended K=2 partition: the smaller group still requires its own B-type fleet because trips may not be re-assigned across groups, and exactly one relay sortie (R01M1) serves areas of both groups, which forces the third relay airframe and explains the replication term. The lexicographic key confirms this: K=2 needs one unique cross-group relay

TABLE X
EXTERNAL CLUSTERING BENCHMARK FOR PROBLEM 4 UNDER IDENTICAL RESOURCE ACCOUNTING (D: TOTAL PEAK DEMAND; G: INVENTORY GAP; J: BALANCE INDEX, SMALLER IS MORE BALANCED; CROSS: CROSS-GROUP RELAY SORTIES).

Method	K	D	G	J	Cross
K-means (geo)	2	36	9	0.17	2
Ward (geo)	2	37	9	0.27	3
geo+demand clustering†	2	29	4	0.76	0
SC-MAE-RG (proposed)	2	**31**	**5**	0.61	**1**
K-means (geo)	3	43	15	0.23	3
Ward (geo)	3	40	12	0.46	3
geo+demand clustering†	3	36–39	9–11	0.54–0.67	2–3
SC-MAE-RG (proposed)	3	**38**	**10**	0.44	3

†The demand-feature clusterings isolate S001 into a singleton group (1+14; 1+7+7 / 1+5+9); this satisfies the shared-sortie coupling (T18 links S011 and S008, which stay together) but violates the three-area operational grouping rule, under which these rows are inadmissible.

sortie and one extra replica, against three of each for $K=3$; no evaluated candidate satisfying the three-area rule achieves fewer cross-group relays for either K, so the comparison is decided on scale and gap. The single-hover baseline of Fig. 3(b) is evaluated under exactly the same link budget, airframe inventory, per-sortie energy model, return-SOC floor and two-stage charging rule as MHPRS, so the coverage gap reflects the mode under the tested implementations rather than more generous resources; the baseline is a one-hover-per-sortie heuristic, not a proven optimum of that mode.

The resource-directed enumeration is benchmarked against generic unsupervised clustering: K-means (100 restarts) and Ward agglomerative clustering on standardized coordinates (geo) and on coordinates plus boxes–mass–volume–first-batch demand (geo+demand). Each clustering output receives the identical S008–S011 feasibility repair (shared sortie T18); all partitions are scored by the same peak-concurrency accounting and rows violating the three-area rule are marked † and excluded from adoption. Table X reports all partitions. For $K=2$, SC-MAE-RG attains the smallest scale and gap ($31/5$) among the evaluated partitions satisfying the shared-sortie coupling and the three-area rule, against $36/9$ and $37/9$ for the geo clusterings, and $38/10$ for $K=3$ against $43/15$ and $40/12$. The geo clusterings obtain a smaller (better) balance index J but at the price of more cross-group relay sorties and replicas, so the comparison is a trade-off among resource scale, inventory gap and cross-group relay redundancy rather than a strict dominance. The demand-feature variants attain a lower D/G ($29/4$ for $K=2$) by isolating S001 (154 kg, $1.9\times$ the next-largest area) into a singleton group (1+14; 1+7+7 and 1+5+9 for $K=3$). These outputs satisfy the shared-sortie coupling—T18, the only multi-area sortie, links S011 and S008, which remain together, and S001's trips are single-area—so they are feasible under the one-area rule and attain the smallest D/G among the evaluated candidates; they are inadmissible only under the three-area rule and are reported for completeness rather than adopted. Generic clustering thus minimizes feature dispersion without perceiving peak concurrency or the non-splittable shared-sortie coupling.

TABLE XI
PEAK CONCURRENT RESOURCE DEMAND OF THE RECOMMENDED $K=2$ PARTITION, AGAINST THE AVAILABLE INVENTORY. G1/G2 = TASK GROUPS; THE GAP COLUMN IS $\max(0, \mathrm{G1}+\mathrm{G2}-\mathrm{inventory})$ PER RESOURCE CLASS ($D=31$, $G=5$).

Resource class	G1	G2	Total	Inventory	Gap
A airframes	2	2	4	4	0
A batteries	2	2	4	6	0
B airframes	2	2	4	2	2
B batteries	3	3	6	4	2
C airframes	2	0	2	2	0
C batteries	4	0	4	4	0
Relay airframes	2	1	3	2	1
Relay energy components	3	1	4	6	0

TABLE XII
TOP FIVE SERVICE AREAS OF THE ENTROPY-WEIGHTED TOPSIS DELIVERY-PRESSURE RANKING (CLOSENESS COEFFICIENT; LARGER MEANS HIGHER DELIVERY PRESSURE).

Area	Mass (kg)	Volume (m^3)	Distance (m)	Closeness	Rank
S001	154	0.394	3063	0.895	1
S003	81	0.211	7261	0.441	2
S002	81	0.211	7435	0.438	3
S004	59	0.156	7746	0.274	4
S006	59	0.156	3181	0.268	5

D. Discussion

The partitioning stage converts the coupled schedule into an actionable decision: the recommended $K=2$ plan has a quantified gap of five resource units (two B-type airframes, two B-type batteries and one relay airframe). The outputs map onto field operations: executable per-trip load and route sheets, four relay sorties over seven service windows verified point-by-point at 1-s resolution, DEM-checked cruise altitudes, and a procurement list naming the five missing units; the zero-replication control shows cross-group relay sharing, not vehicle count, to be binding.

Two threats to validity deserve emphasis. First, the energy model is validated only against frozen computational artifacts: the reconstruction reproduces the relay-required count within 0.3% and the relay energy within 0.1%, but the hover points of two of the seven service windows are re-derived from the DEM grid rather than read from the frozen log, so this is a reconstruction, not an independent flight measurement, and no hardware test has been performed; wind, payload-drag and battery-aging effects could shift the absolute figures. The reconstructed timeline also assumes an aggregate 37 s climb allowance for A/B departures (those samples remain directly connected), a raised corridor on S007 legs, and a 6 m clearance with bilinear interpolation in the direct-link test; the relay-coverage test uses the raw DEM profile, so its 100% figure is tied to the reconstruction convention; at 6 m, one reconstructed hover point leaves a service window uncovered. A local 256 m adjustment restores sampled coverage of that window (arrival slack 582 s, return SOC 0.41, and sortie energy 1.2% above the frozen value). The full scheduling pipeline was not rerun under this rule. Second, relay demand is identified on a five-second search grid, which underestimates the relay-required population relative to the one-second verification grid; the 11,713-sample coverage claim therefore holds only on

the one-second grid under the stated link model, and the candidate search may miss short-duration blockage events. Further limitations are that link feasibility is judged by a binary DEM line-of-sight test with a fixed occlusion loss, and that the environment is deterministic and does not model weather or equipment faults. For Problem 2, the joint batching–routing–scheduling optimum is not searched, so the 65.25 kWh main plan remains the best of the enumerated candidate structures under the urgency-first scheduler rather than a proven joint optimum. Because the parameters are data-driven, the chain transfers to other mountainous settings, but the reported fleet configuration and coverage figures are instance-specific.

Transferring the chain to a new site requires re-running rather than re-designing: recompute the safe-payload table from a regional DEM, enter the available fleet into the parameter tables, re-identify the relay-required population, and run the four stages in order; the tour enumeration and the per-second verification are embarrassingly parallel.

V. Conclusion

This paper built a four-stage "transport scheduling–communication support–resource allocation" framework on one consistent physical model, in which the core stage constructs a relay schedule once the delivery plan is fixed. The exact batching model returns an 18-trip plan at 59.24 kWh with a verified optimality benchmark; the heterogeneous routing model delivers all 80 boxes, satisfies all 31 hard deadlines and recommends a 20-trip plan at 65.25 kWh; and the relay stage covers all 11,713 sampled relay-required points with two airframes and four sorties at 5.43 kWh (joint 70.68 kWh). The partitioning stage then quantifies the resource gap and produces an explicit augmentation list. The sensitivity studies delineate the envelope: batching tolerates return margins up to 35% (failing at 40%), relay demand is governed by the receiving threshold rather than the terrain clearance, and among the tested implementations, only MHPRS achieves full sampled coverage on the 1-s grid with two relay airframes.

Further work should replace the one-second grid with event-driven continuous-time link checking, add wind, weather, stochastic demand and multi-depot sharing, couple scheduling with batching in one joint model, and report acceptance rates on end-to-end geographic families.

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Data availability: Code, data and verification pipeline: `https://github.com/vscode-aiyaya/uav-flood-transport` (archival DOI upon acceptance).

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